TOR AMPLITUDE IN A BOUNDED INTERVAL TOGETHER WITH (A-1)-PSEUDO-COHERENCE FORCES PERFECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the localisation to a strictly perfect witness, the truncation, the identification of the cone with the shifted kernel, the computation of the vanishing Tor_1 of the cokernel from tor-amplitude, the flatness criterion and the final passage to a locally finite free summand.

Editorial note. The label lemma-perfect-precise names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Tor amplitude in an interval and (a-1)-pseudo-coherence. Same chapter, the definitions of tor amplitude in [a, b] and of m-pseudo-coherence used as the two hypotheses of the statement. Supplies both hypotheses and the strictly perfect witness alpha whose truncation is used in the proof.
- Flatness criterion for modules with vanishing first Tor. Same chapter, the lemma concluding flatness of an O-module from the vanishing of Tor_1 against all modules. Supplies flatness of the cokernel E' = Coker(E^{a-1} -> E^a) from the computed vanishing.
- Locally finitely presented flat modules are direct summands of finite free modules. Modules on Sites, the lemma describing flat modules of finite presentation over a ringed site. Supplies the covering over whose members E' becomes a direct summand of a finite free module.

Source reference. Original lemma 00HL: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3360–3360. Prerequisite pointer invoked at proof line(s) 12804; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08FD: sources/stacks-chow/tags.txt, source lines 10706–10706. Prerequisite pointer invoked at proof line(s) 12808; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let E be an object of $D(\mathcal{O})$. Let $a \leq b$ be integers. If E has tor amplitude in $[a, b]$ and is $(a-1)$-pseudo-coherent, then E is perfect.*

Proof. Let U be an object of $\mathcal{C}$. After replacing U by the members of a covering and $\mathcal{C}$ by the localization $\mathcal{C}/U$ we may assume there exists a strictly perfect complex $\mathcal{E}^\bullet$ and a map $\alpha : \mathcal{E}^\bullet \to E$ such that $H^i(\alpha)$ is an isomorphism for $i \geq a$. We may and do replace $\mathcal{E}^\bullet$ by $\sigma_{\geq a-1}\mathcal{E}^\bullet$. Choose a distinguished triangle

$$\mathcal{E}^\bullet \to E \to C \to \mathcal{E}^\bullet[1]$$

From the vanishing of cohomology sheaves of E and $\mathcal{E}^\bullet$ and the assumption on α we obtain $C \cong \mathcal{K}[2-a]$ with $\mathcal{K} = \mathrm{Ker}(\mathcal{E}^{a-1} \to \mathcal{E}^a)$. Let $\mathcal{F}$ be an $\mathcal{O}$-module. Applying $- \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}$ the assumption that E has tor amplitude in $[a, b]$ implies $\mathcal{K} \otimes_{\mathcal{O}} \mathcal{F} \to \mathcal{E}^{a-1} \otimes_{\mathcal{O}} \mathcal{F}$ has image $\mathrm{Ker}(\mathcal{E}^{a-1} \otimes_{\mathcal{O}} \mathcal{F} \to \mathcal{E}^a \otimes_{\mathcal{O}} \mathcal{F})$. It follows that $\mathrm{Tor}_1^{\mathcal{O}}(\mathcal{E}', \mathcal{F}) = 0$ where $\mathcal{E}' = \mathrm{Coker}(\mathcal{E}^{a-1} \to \mathcal{E}^a)$. Hence $\mathcal{E}'$ is flat (Lemma 00HL). Thus there exists a covering $\{U_i \to U\}$ such that $\mathcal{E}'|_{U_i}$ is a direct summand of a finite free module by Modules on Sites, Lemma 08FD. Thus the complex

$$\mathcal{E}'|_{U_i} \to \mathcal{E}^{a+1}|_{U_i} \to \ldots \to \mathcal{E}^b|_{U_i}$$

is quasi-isomorphic to $E|_{U_i}$ and E is perfect. □